SAI SREEKAR SIDDULA

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SUMMARY

Senior Software Engineer with **6+ years** building large-scale, production React/TypeScript applications. Led development of the most complex app on a national retail healthcare platform used daily across **thousands of locations**: long-running stateful workflows, real-time concurrency control, browser-based hardware integration, and one codebase serving desktop workstations and handheld devices. Owns ambiguous, cross-team problems from system design through progressive rollout and production observability.

WORK EXPERIENCE

Branding Brand | CVS Health **February 2025 – Present**
Senior Software Engineer

- Led development of the platform's most complex app, a **dual-target React SPA** (desktop workstation and Zebra Android handheld from one codebase, ~90 platform-specific screens) running **30+ long-lived, multi-step workflows** with branching state, resume-after-interruption, and dependencies on backend, hardware, and other users.
- Designed a **scheduling engine** for multi-dose vaccine series, where moving one appointment reshuffles the next: linked time-slot holds, minimum-interval rules with cascading invalidation, permission-gated overrides, and race-safe parallel booking calls.
- Built **frontend-coordinated record locking** so two staff can never edit the same record at once: all-or-nothing acquisition, partial release, recovery across page reloads, per-device release, and distinct conflict screens for another user versus your own other device.
- Integrated four kinds of **hardware into the browser**: a custom GS1-128 barcode parser extracting drug identifiers, dual concurrent camera streams for OCR and photo capture, label printers, and biometric sign-in via native SDK and WebSocket adapters.
- Enforced runtime type safety at every network boundary: **60+ endpoints** wrapped in **Zod**-validated contracts with normalizers mapping raw API data to typed view models; partnered with backend architects on **HL7 / FHIR** data contracts.
- Designed recovery paths for **69 distinct API failure scenarios** (validation, business-rule, concurrency, mid-workflow), each reproducible in tests; workflows survive hard navigation and context changes, releasing held slots and locks.
- Shipped every major feature behind **9 runtime flags** and 20+ per-location config keys, enabling progressive rollout across thousands of pharmacies with differing regulatory requirements without redeploying.
- Cut redundant re-renders to improve page performance **~30%** in high-volume stores, and built keyboard-driven navigation that lets technicians process prescriptions **30% faster**.
- Built features across the platform's other two apps in a shared **~500K-LOC Nx monorepo**: **prescriber-facing** (provider search, identity lookup, verification) and **operations** (batch verification, work queues, role-based routing); instrumented production with **OpenTelemetry** tracing and real-user monitoring.

Ductus Inc. | Crown Castle Inc. **May 2020 – August 2024**
Software Engineer II *June 2021 – August 2024*

- Led frontend delivery for a telecom client's legacy **modernization** across 10+ React/Angular projects, from requirements and **Figma** wireframes to deployment, backed by **Python** microservices behind a **gRPC** backend-for-frontend layer.
- Redesigned the frontend architecture to cut delivery turnaround **33%**, shipping a large app in 4 months instead of 6.
- Designed and drove adoption of a **Material UI** component library across 7 projects, cutting new-feature development time **30%**.
- Established a testing strategy (Testing Library, Jest) that lifted coverage to **90%** and cut reported bugs **30%**; added a Swagger mock server, GitHub Copilot, and Figma dev mode to cut development time another **20%**.

Software Engineer Co-Op *May 2020 – December 2020*

- Built a CLI automating **CloudFormation** template generation that cut AWS deployment time **40%**, and a suite of **VanillaJS** web components adopted across 5 client projects for cross-framework reuse.

SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java
Frontend: React 19, Angular, Next.js, TanStack Query, Redux, GraphQL, RxJS, Zod, Tailwind, Material UI, D3.js, Vite
Backend & Data: Node, Express, gRPC, REST, WebSockets, JPA, MongoDB, MySQL, HL7 / FHIR, OAuth
Testing & Observability: Vitest, Jest, React Testing Library, Playwright, Cypress, msw, pytest, OpenTelemetry
Platform & AI: Nx, Bun, Docker, GCP, Azure DevOps, Vercel, Biome, Figma; LLM integration, agentic workflows

EDUCATION

Northeastern University **September 2019 – May 2021**
M.S., Computer Systems Engineering · GPA 4.0

Shiv Nadar University **August 2014 – May 2018**
B.Tech., Electronics & Communication Engineering, Minor in Computer Science · GPA 3.7